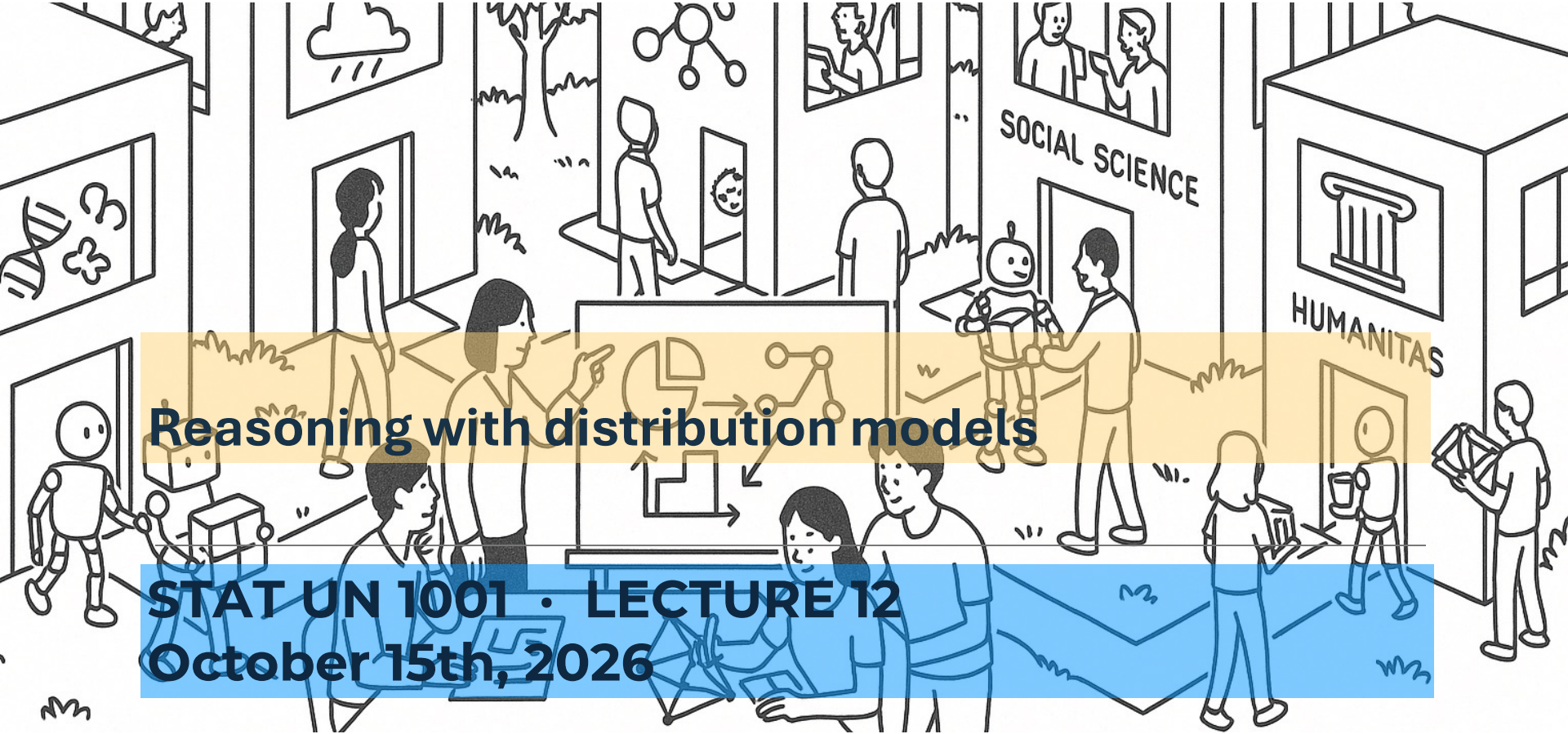


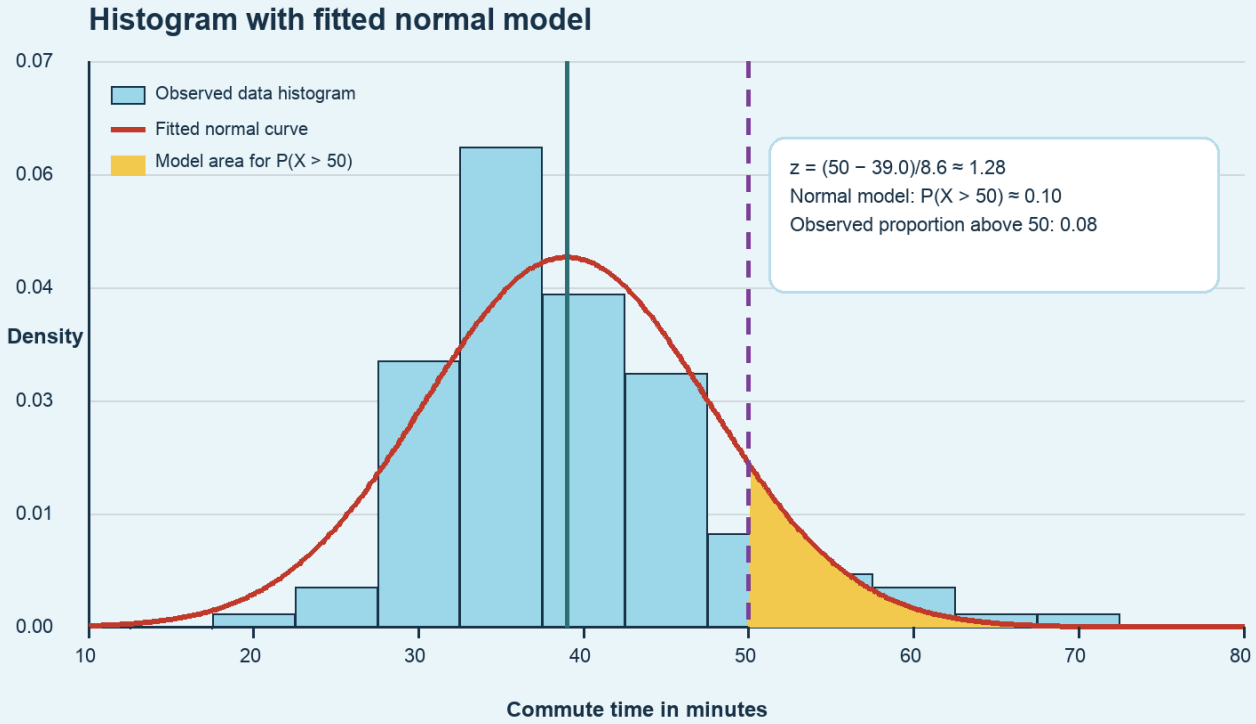
Reasoning with distribution models

STAT UN 1001 · LECTURE 12
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Distribution model thinking

1	**Observed data**	a histogram, a table, a sample proportion
2	**Model**	an idealized chance process with probabilities
3	**Event**	a set of values we care about
4	**Probability**	area under a curve, or a formula
5	**Check**	does the model fit the context well enough?

A histogram with a fitted normal curve



What does the histogram show?

Curve: a distribution model

Guided example 1: commute above 50

Part	Fill in together
Chance process	randomly sample one commute from this setting
Random variable	X = commute time in minutes
Event	X > 50
Model	Normal with mean about 39.0 and SD about 8.6

Interpret first: “We want the chance a randomly sampled commute is longer than 50 minutes.”

Standardizing the cutoff

Why standardize? It turns the original scale into standard-deviation units.

Step	Computation	Meaning
subtract mean	$50 - 39.0 = 11.0$	11 minutes above average
divide by SD	$11.0 / 8.6 \approx 1.28$	1.28 SDs above average
probability sentence	$P(X > 50) = P(Z > 1.28)$	right-tail area

Ask students before showing the table: should this be a small, medium, or large right-tail probability?

From z-score to probability

$$P(X > 50) = P(Z > 1.28) \approx 0.10$$

Quantity	Value	How to read it
Observed sample proportion above 50	about 0.08	what happened in this sample
Normal-model probability above 50	about 0.10	what the fitted model predicts

Guided example 2: compare two commutes

Student	Commute	Context mean	Context SD	Question
A	50 min	39 min	8.6 min	How unusual?
B	35 min	25 min	5 min	How unusual?

PAIR DISCUSSION

Which commute is longer in minutes? Which commute is more unusual relative to its context?

Key distinction: raw value and standardized value answer different questions.

Standardized comparison

Student	z-score	Interpretation
A	$(50 - 39) / 8.6 \approx 1.28$	1.28 SDs above the city mean
B	$(35 - 25) / 5 = 2.00$	2.00 SDs above the city mean

Student B is more unusual relative to context, even though Student A has the longer commute.

Binomial model: what it is for

A binomial model counts the number of successes in a fixed number of yes/no trials.

Condition	Meaning	Phone-message example
Fixed n	same number of trials	5 messages
Two outcomes	success or failure	typo or no typo
Same p	same success probability	p = 0.30 each message
Independent	one trial does not change another	assume messages are independent

Guided example 3: exactly two typos

SETUP

Suppose 5 independent messages each have probability 0.30 of containing a typo.

Symbol	Meaning
X	number of messages with typos
n	5 messages
p	0.30 per message
Event	X = 2

Ask first: what would one specific order look like? Example: typo, typo, no, no, no.

Exact binomial probability

Piece	Value	Why
choose the two typo slots	C(5, 2) = 10	10 possible positions
two typo messages	(0.30)^2	two successes
three no-typo messages	(0.70)^3	three failures

$$P(X = 2) = C(5, 2)(0.30)^2(0.70)^3 \approx 0.309$$

Interpretation: across many groups of five messages, about 31% of groups would have exactly two typo messages.

Binomial mean and standard deviation

Quantity	Formula	For n = 5, p = 0.30	Meaning
Mean	np	5(0.30) = 1.5	long-run average count
SD	sqrt(np(1−p))	sqrt(5×0.30×0.70) ≈ 1.02	typical variation around mean

The mean is not a promised count for one set of five. It is the average count over repeated sets of five.

Distribution model reasoning

1	**Chance process**	what random process are we modeling?
2	**Random variable**	what number are we tracking?
3	**Event**	what set of values matters?
4	**Model**	normal curve or binomial count?
5	**Computation**	area, z-score, or binomial formula?
6	**Interpretation**	what does the probability mean in context?

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